Low-Level Design - LLD

1. Introduction

The Low-Level Design (LLD) provides a detailed breakdown of how the RAG-based Customer Support Assistant is implemented internally. While the High-Level Design (HLD) outlines the architecture and conceptual flow, the LLD focuses on **modules, data structures, workflows, routing logic, HITL integration, and error handling**.

This document ensures that engineers can understand the exact implementation details, replicate the system, and extend it for future enhancements.

2. Module-Level Design

The system is modular, with each Python file responsible for a specific function.

2.1 ingest.py

- **Purpose:** Load the PDF knowledge base, split into chunks, generate embeddings, and store them in ChromaDB.
- **Steps:**
    1. Load PDF using PyPDFLoader.
    2. Split text into chunks (1000 characters, 200 overlap).
    3. Generate embeddings using HuggingFace MiniLM-L6-v2.
    4. Store embeddings in ChromaDB with persistence.

2.2 query.py

- **Purpose:** Retrieve relevant chunks for a given query.
- **Steps:**
    1. Convert query into embedding.
    2. Perform similarity search in ChromaDB.
    3. Return top-k chunks.

2.3 llm_groq.py

- **Purpose:** Generate contextual answers using Groq LLM.
- **Steps:**
    1. Accept query and retrieved chunks.
    2. Construct prompt with context.
    3. Call Groq API (model: llama3-70b-8192).
    4. Return generated answer.

2.4 workflow.py

- **Purpose:** Orchestrate query processing using LangGraph.
- **Steps:**
    1. Input node receives query.
    2. Router node decides whether AI can answer or escalate.
    3. Answer node generates response via Groq.
    4. HITL node escalates to human if needed.
    5. Output node formats final response.

2.5 hitl.py

- **Purpose:** Handle Human-in-the-Loop escalation.
- **Steps:**
    1. Log escalated query.
    2. Notify human agent.
    3. Store human-provided answer.
    4. Return answer to user.

2.6 app.py

- **Purpose:** Provide Streamlit UI for user interaction.
- **Features:**

    - Text input for queries.
    - Display answers.
    - Show escalation status.

2.7 main.py

- **Purpose:** Unified entry point for CLI interaction.
- **Features:**
    - Accept queries.
    - Call workflow.
    - Display answers.

3. Data Structures

3.1 Document Representation

- Raw text extracted from PDF.
- Metadata: page number, section.
- The pdf data belongs to Customer Support of People Partnership Solutions

3.2 Chunk Format

```
{
  "id": "chunk_001",
  "page": 10,
  "text": "Priority Level & Initial Response Times..."
}
```

3.3 Embedding Structure

- Vector of 384 dimensions (MiniLM-L6-v2).
- Stored in ChromaDB.

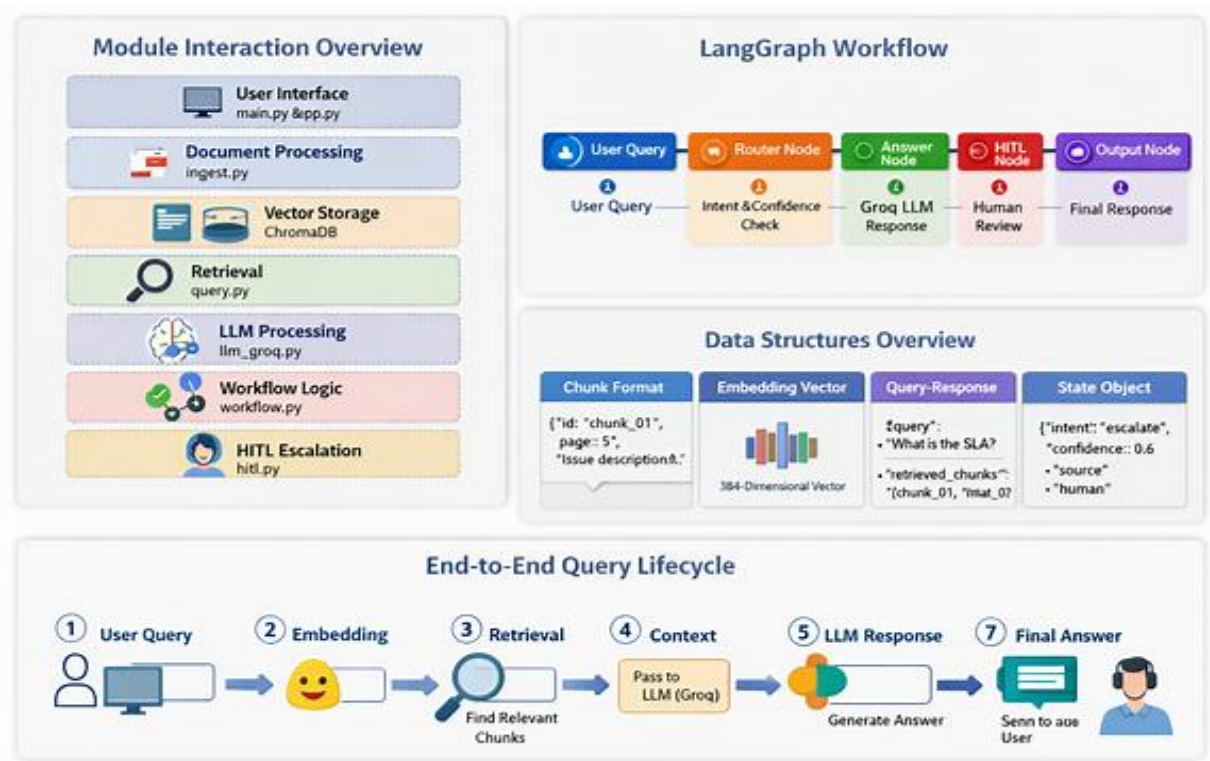
3.4 Query-Response Schema

```
{
  "query": "What is the SLA for Priority 1 incidents?",
  "retrieved_chunks": ["chunk_001", "chunk_002"],
  "answer": "Priority 1 incidents have a target response time of 1 hour."
}
```

3.5 LangGraph State Object

```
{
  "intent": "answerable",
  "confidence": 0.85,
  "source": "llm",
  "final_response": "Priority 1 incidents have a target response time of 1 hour."
}
```

4. Workflow Design (LangGraph)



Nodes

- **Input Node:** Accepts query.
- **Router Node:** Determines intent and confidence.
- **Answer Node:** Calls Groq LLM.
- **HITL Node:** Escalates to human agent.
- **Output Node:** Formats final response.

Edges

- Input → Router → Answer → Output.
- Input → Router → HITL → Output.

State Transitions

- Confidence ≥ 0.7 → Answer Node.
- Confidence < 0.7 → HITL Node.

5. Conditional Routing Logic

Routing is based on **confidence scores** and **intent detection**.

- **Answerable Queries:** Routed to Groq LLM.
- **Low Confidence Queries:** Routed to HITL.
- **Complex Queries:** Routed to HITL.
- **Missing Context:** Routed to HITL.

6. HITL Design

Escalation Triggers

- Confidence < 0.7.
- No relevant chunks found.

- Query outside knowledge base.

Human Agent Role

- Review escalated query.
- Provide factual answer.
- Update knowledge base if needed.

Integration

- HITL answers are logged.
- Future queries benefit from updated knowledge.

7. API / Interface Design

CLI Interface

- Input: text query.
- Output: text answer.

Streamlit UI

- Input: text box.
- Output: formatted answer.
- Status: AI or HITL.

8. Error Handling

- **Missing PDF:** Prompt user to upload.
- **No Chunks Found:** Re-ingest document.
- **LLM Failure:** Escalate to HITL.
- **Network Error:** Retry or escalate.

Conclusion

The Low-Level Design provides a detailed blueprint for implementing the RAG-based Customer Support Assistant. By defining modules, data structures, workflows, routing logic, HITL integration, and error handling, the LLD ensures that the system is robust, scalable, and reliable.